

Measurement

Module 7 Test

Grade 9 • MTH1W

Time: 60 minutes • Total: 70 marks

Name: _____

Date: _____

Instructions: Show all work for full marks. Marks for each question are shown in brackets. Use $\pi \approx 3.14$ and round π -answers to **1 d.p.** Calculators are permitted unless your teacher states otherwise.

Section A — Knowledge & Understanding

1. Find the area of a rectangle 7 cm by 6 cm. **[1]**

2. Find the perimeter of a rectangle 8 cm by 3 cm. **[1]**

3. Find the area of a square with side 9 cm. **[1]**

4. Find the area of a triangle with base 8 cm and height 5 cm. **[1]**

5. Find the volume of a cube with edge 3 cm. **[1]**

6. State the formula for the volume of a cylinder. **[1]**

7. A circle has radius 10 cm. Find its circumference ($\pi \approx 3.14$, 1 d.p.). **[1]**

8. State the formula for the area of a circle. **[1]**

9. Find the perimeter of a square with side 12 cm. [1]

10. Find the area of a parallelogram with base 11 cm and height 4 cm. [2]

11. Find the area of a trapezoid with parallel sides 8 cm and 4 cm and height 6 cm. [2]

12. A circle has radius 4 cm. Find its area ($\pi \approx 3.14$, 1 d.p.). [2]

13. Find the volume of a rectangular prism 5 cm by 4 cm by 6 cm. [2]

14. Find the surface area of a cube with edge 7 cm. [2]

15. A cylinder has radius 3 cm and height 5 cm. Find its volume ($\pi \approx 3.14$, 1 d.p.). [2]

Section B — Application

16. A triangle has base 14 cm and height 9 cm. Find its area. [2]
17. A rectangular prism is 8 cm by 5 cm by 4 cm. Find its surface area. [2]

18. A cylinder has radius 6 cm and height 10 cm. Find its volume ($\pi \approx 3.14$, 1 d.p.). **[2]**
19. A cone has radius 3 cm and height 8 cm. Find its volume ($\pi \approx 3.14$, 1 d.p.). **[2]**
20. A sphere has radius 7 cm. Find its surface area ($\pi \approx 3.14$, 1 d.p.). **[2]**
21. A square-based pyramid has base edge 5 cm and height 12 cm. Find its volume. **[2]**
22. A cylinder has diameter 8 cm and height 9 cm. Find its surface area ($\pi \approx 3.14$, 1 d.p.). **[3]**
23. A composite figure is a 14 cm by 8 cm rectangle with a semicircle of diameter 8 cm on one end. Find the total area ($\pi \approx 3.14$, 1 d.p.). **[3]**

24. A sphere has radius 5 cm. Find its volume ($\pi \approx 3.14$, 1 d.p.). **[3]**

Section C — Thinking

25. A cylindrical water tank has radius 2 m and height 3 m. Find its volume, then how many litres it holds ($1 \text{ m}^3 = 1000 \text{ L}$). ($\pi \approx 3.14$, 1 d.p.) **[4]**
26. A composite solid is a cube of edge 5 cm with a square-based pyramid (base edge 5 cm, height 9 cm) sitting on top. Find the total volume. **[4]**

27. A cylinder has radius 5 cm and height 20 cm. It is melted and recast into spheres of radius 5 cm. How many complete spheres can be made? (Very hard.) ($\pi \approx 3.14$, 1 d.p.) **[5]**
28. A cone has radius 9 cm and height 12 cm. Find its slant height, then its total surface area. (Very hard.) ($\pi \approx 3.14$, 1 d.p.) **[5]**

29. A grain silo is a cylinder of radius 4 m and height 9 m topped by a cone of radius 4 m and height 3 m. Find the total volume of the silo. (Very hard.) ($\pi \approx 3.14$, 1 d.p.) **[6]**

Section D — Communication

30. A cube and a sphere have exactly the same volume. Explain, with reasoning, which one has the smaller surface area, and state the general optimization principle this illustrates. (Very hard.) **[4]**

— END OF EXAM —